

# StaleLine: Market Structure

Andrew deliverable — market\_structure.md | Core test: Kalshi vs. Polymarket.com + Polymarket US; CME is background

## What matters for us

The core latency test now compares **Kalshi, Polymarket.com (International), and Polymarket US**. All three are electronic order-book markets, but their participant pools, access paths, rules, and sports-order mechanics differ. Polymarket.com has an explicit sports taker delay and clears sports resting orders at game start; Polymarket US uses its own CFTC-regulated order book and current public rules do not specify the same sports-specific delay or start-time wipe. CME remains background evidence that professional electronic market makers can matter for price discovery, not a core execution venue.

## Kalshi

### Launch date and regulator

KalshiEX LLC received CFTC designation as a **Designated Contract Market (DCM) in November 2020** and opened its public beta in **July 2021**. The relevant football family is KXNFLGAME. *Sources:* [Kalshi designation](#); [public beta](#); [CFTC DCM list](#).

### Contract size, price increment, trading hours

Kalshi NFL winner contracts are binary contracts that pay **\$1** for the winning outcome; current NFL market pages quote prices in cents and resolve ties according to the specific market rules. For implementation, the price increment should be read from Kalshi's market metadata (`price_ranges / step`) rather than hard-coded across every market. Kalshi is open **24/7 except scheduled maintenance, currently Thursday 3:00–5:00 AM ET**. *Sources:* [example KXNFLGAME market](#); [Kalshi API](#); [trading hours](#).

### Who trades it and how they reach it

Kalshi serves direct users through its web/app interface and API and also reaches customers through broker distribution such as Webull. The exchange explicitly supports professional liquidity provision through a **Market Maker Program**; therefore, the venue is not purely retail. *Sources:* [participant structure](#); [market makers](#).

### How orders are matched and how fast prices update

Kalshi uses an electronic limit order book with resting bids/offers and maker/taker execution. For StaleLine, the actionable price is the executable best bid or ask, not the last trade. Kalshi does not publish a guaranteed event-to-quote reaction time; our latency estimate comes from the synchronized book data. *Sources:* [order book](#); [taking available liquidity](#).

### Published volume / open interest for a typical NFL game

Kalshi publishes market/event volume on NFL pages, but volume varies materially by game and page-level figures can include related markets. For the deliverable and note, the defensible “typical” figure is the **median volume/open interest of the exact KXNFLGAME markets in our matched sample**; public NFL pages are context, not the estimator. *Source:* [Kalshi NFL hub](#).

## Polymarket.com (International)

### Launch date and regulatory status

Polymarket's international prediction-market product dates to **2020**. It is a separate product from Polymarket US; current Polymarket guidance says **U.S. residents cannot trade on polymarket.com** and must use the U.S. product. CFTC DCM status applies to Polymarket US, not the international website. *Sources:* [Polymarket transparency](#); [International vs. U.S. products](#).

### Contract size, price increment, trading hours

Outcome shares trade between 0 and 1 and a winning share redeems for **\$1**. Tick size is market-specific in the order-book metadata and should be read from the live market rather than assumed. The international platform operates continuously subject to maintenance and geographic restrictions. *Sources:* [contract mechanics](#); [geographic restrictions](#).

### Who trades it and how they reach it

International users trade through polymarket.com and its APIs. The venue has retail users plus active liquidity providers; Polymarket explicitly pays **Liquidity Rewards** and **Maker Rebates** to incentivize competitive quoting. *Sources:* [Liquidity Rewards](#); [Maker Rebates](#).

### How orders are matched and how fast prices update

Polymarket.com uses an electronic order book. Limit orders can rest and partially fill; the website's displayed probability is normally the **bid/ask midpoint**, and if the spread exceeds 10 cents it displays the last trade instead. For sports, two rules directly matter to StaleLine: **all outstanding sports limit orders are automatically cancelled at official game start**, and **marketable sports orders have a 3-second placement delay**; Polymarket also states that a 1-second taker delay is being tested on NBA and MLB markets. These mechanics can create measured latency independently of trader information processing. *Sources:* [displayed prices](#); [sports order rules and delay](#).

### Published volume / open interest for a typical NFL game

Polymarket publishes market-level NFL volume, but activity varies widely by matchup and game state. Our “typical” statistic should therefore be the **median volume of the exact matched NFL moneyline markets in our sample**. *Source:*

[Polymarket NFL markets.](#)

## Polymarket US

### Launch date and regulator

Polymarket US is operated by **QCX LLC d/b/a Polymarket US**, a CFTC-regulated DCM. QCX received DCM designation on **July 9, 2025**; Polymarket’s transparency page states the U.S. exchange has been **fully launched since May 2026**. *Sources:* [CFTC DCM list](#); [Polymarket transparency](#); [PMUS regulatory page](#).

### Contract size, price increment, trading hours

Polymarket US Athletic Outcome Contracts have **\$1 notional size**. Product terms permit prices from \$0.001 to \$0.999 and specify a **minimum tick between \$0.001 and \$0.01**, with the contract-specific tick set by the Exchange. The exchange is open **24/7**, with scheduled maintenance currently **Thursday 6:00–8:00 AM ET**. *Sources:* [Athletic Outcome Contract terms](#); [operating hours](#).

### Who trades it and how they reach it

Polymarket US admits individual and legal-entity Participants and supports access through its U.S. app / independent software vendors as well as institutional channels. Its rulebook permits designated **Market Makers**, and its 2026 Market Maker Program requires participating firms to maintain two-sided markets within defined spreads and depths. *Sources:* [PMUS Rulebook](#); [Market Maker Program](#); [institutional/FIX/data contacts](#).

### How orders are matched and how fast prices update

The PMUS Direct System matches orders electronically by **price priority, then time priority**. Standing limit orders can rest until filled, cancelled, expired, or replaced; market-to-limit and fill-and-kill instructions are also permitted. The current PMUS Rulebook and Athletic Outcome Contract terms do **not specify the polymarket.com sports rule that automatically wipes the book at game start or imposes a 3-second taker delay**. Therefore the core test must treat PMUS as a separate venue and should not copy the international delay into its simulation unless a contract-specific notice or interface documents it. *Sources:* [PMUS Rulebook, Rules 5.4–5.5](#); [AOC terms](#).

### Published volume / open interest for a typical NFL game

Polymarket US publishes **daily market reports and time-and-sales reports**, allowing contract-level activity to be reconstructed. It does not publish one venue-defined “typical NFL game” statistic; use the median volume/open interest from the exact PMUS NFL contracts in our matched sample. *Sources:* [daily market reports](#); [time and sales](#).

## CME sports event contracts — background only

CME launched its swap-based sports event contracts on **December 5, 2025** on CME Globex. CME is a CFTC-regulated DCM and has a formal event-contract market-maker program for proprietary trading firms. This remains useful institutional precedent for rapid electronic price discovery, but CME is no longer part of the core StaleLine test. *Sources:* [CME launch notice](#); [market-maker program](#).

## Core-test comparison

| Feature                                   | Kalshi                                                 | Polymarket.com                                         | Polymarket US                                                                        |
|-------------------------------------------|--------------------------------------------------------|--------------------------------------------------------|--------------------------------------------------------------------------------------|
| <b>Regulatory/access structure</b>        | CFTC DCM; direct app/web/API + brokers                 | International platform; U.S. users cannot trade        | CFTC DCM; U.S. app/ISVs/direct institutional access                                  |
| <b>Payoff</b>                             | \$1 / \$0                                              | winning share \$1                                      | \$1 / \$0                                                                            |
| <b>Professional liquidity</b>             | Market Maker Program                                   | Liquidity Rewards + Maker Rebates                      | formal Market Maker Program                                                          |
| <b>Matching</b>                           | electronic maker/taker book                            | electronic order book                                  | price-time priority electronic book                                                  |
| <b>Sports-specific execution mechanic</b> | none publicly specified beyond normal order-book rules | game-start book wipe + 3-second marketable-order delay | no equivalent sports-specific delay/wipe stated in current public Rulebook/AOC terms |
| <b>Price input for StaleLine</b>          | executable bid/ask quotes                              | executable bid/ask quotes, not headline midpoint       | executable bid/ask quotes                                                            |

## Why one venue might reprice first

### Judge-ready two sentences:

“Kalshi, Polymarket.com, and Polymarket US all have professional liquidity providers, but they are separate order books with different participant pools and execution rules, so the same football information can reach executable prices at different speeds. Polymarket.com’s documented sports taker delay is a concrete venue-level friction, while Kalshi and Polymarket US use different access and market-maker structures, so we measure the leader empirically from synchronized executable quotes rather than assuming one venue is always fastest.”